Reg. No. | | | | | | | | | | | | |

BCACACS 103

First Semester B.C.A. Degree Examination, December 2024/January 2025
(SEP) (2024 – 25 Batch Onwards)
DISCRETE MATHEMATICS FOR COMPUTER APPLICATIONS

Time : 3 Hours Max. Marks : 80

Note : *Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit of Part – **B**.*

PART – A

(10×2=20)

1. a) Construct the truth tables for
   i) $\neg p \wedge \neg q$
   ii) $\neg (p \vee q) \wedge R$.

   b) Write the power set of A = {1, 2}.

   c) Represent
   i) $A \cup B$
   ii) $A - B$ using Venn diagram.

   d) Define surjective function. Give an example.

   e) How many different bit strings of length seven are there ?

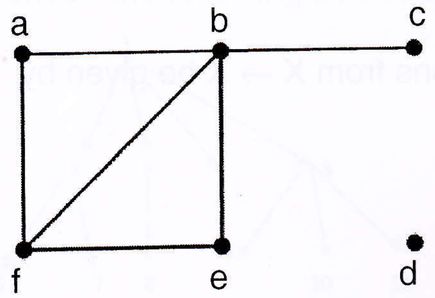
   f) There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there ?

   g) Write the formula to calculate P(n, r).

   h) An urn contains four blue balls and five red balls. What is the probability that a ball chosen at random from the urn is blue ?

   i) What is the greatest common divisor of 24 and 36 ?

   j) Find isolated and pendant node in the graph.



P.T.O.

k) Define complete graph with an example.

l) Design an NFA to accept the set of strings over the alphabet set {0, 1} and ending with two consecutive zeros.

PART – B

Unit – I

2. a) Using truth table, show that $(p \wedge q) \Rightarrow p$ and $p \Rightarrow (p \vee q)$ are both tautologies, where p and q are any two statements.

   b) A = {1, 2, 3}, B = {1, 2, 5, 7, 9}. Write A – B, B – A, A + B, $A \cup B$, $A \cap B$.

   c) Let X = {1, 2, 3, 4} and R = {(x, y)|x > y}. Draw the graph of R and give its matrix. **(5+5+5)**

3. a) Show the following implications using truth table.

      i) $(p \wedge q) \Rightarrow (p \rightarrow q)$

      ii) $p \Rightarrow (q \rightarrow p)$.

   b) A = {1}, B = {a, b}, C = {2, 3} write $A \times B$, $B \times A$, A^2, B^2, $A^2 \times B$.

   c) Let X = {1, 2, 3, ..., 7} and R = {<x, y>|x – y is divisible by 3}, show that R is an equivalence relation. **(5+5+5)**

Unit – II

4. a) i) How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene) ?

      ii) Let S = {1, 2, 3, 4, 5} find the 3-premutation and 3-combination of S.

   b) Let f(x) = x + 2, g(x) = x – 2 and h(x) = 3x for $x \in R$, R is a set of real numbers. Find $f \circ f$, $f \circ g$, $g \circ g$, $g \circ f$ and $f \circ (h \circ g)$.

   c) Show that functions $f(x) = x^3$ and $g(x) = x^{1/3}$ for $x \in R$ are inverse of one another. **(5+5+5)**

5. a) Let X = {2, 3, 6, 12, 24, 36} and the relation $\leq$ be such that $x \leq y$, if x divides y. Draw the Hasse diagram of $(X, \leq)$.

   b) Let x = {1, 2, 3} and f, g, h and s be functions from $X \rightarrow X$ be given by

      f = {<1, 2>, <2, 3>, <3, 1>}

      g = {<1, 2>, <2, 1>, <3, 3>}

      h = {<1, 1>, <2, 2>, <3, 1>}

      s = {<1, 1>, <2, 2>, <3, 3>}

      Find $f \circ g$, $g \circ f$, $s \circ s$, $f \circ h \circ g$ and $f \circ s$.

c) A computer company receives 350 applications from computer graduates for a job planning a line of new Web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business ? **(5+5+5)**

Unit – III

6. a) Show that if n is a positive integer, then $1+2+3+....+n=\frac{n(n+1)}{2}$.

b) A fair coin is flipped three times. Let S be the sample space of the eight possible outcomes, and let X be the random variable that assigns to an outcome the number of heads in this outcome. What is the expected value of X ?

c) Find the greatest common divisor of 414 and 662 using the Euclidean algorithm. **(5+5+5)**

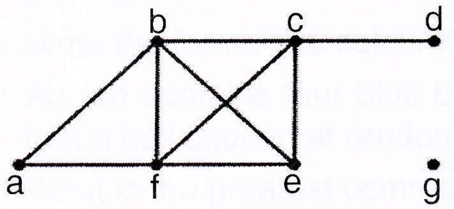
7. a) What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5 ?

b) Use mathematical induction to prove that $2^n < n!$ for every integer n with $n \geq 4$.

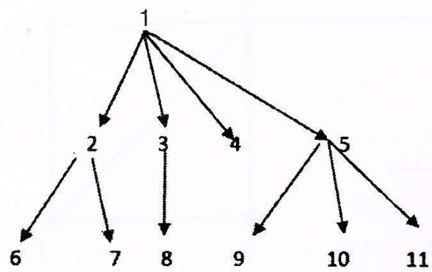
c) Find the prime factorization of 7007. **(5+5+5)**

Unit – IV

8. a) What are the degrees and neighbourhood of the vertices in graph displayed ?

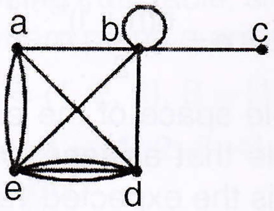


b) Convert the following tree into a binary tree.

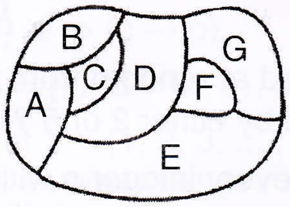


c) Give a state diagram of finite state automata of a devices that determines a modulo 3 counter that receives sequence of 0's, 1's and 2's as input and produces a sequence of 0's, 1's and 2's as a output such that at any instance of the output is equal the modulo 3 sum of digits in the input sequences. **(5+5+5)**

9. a) What are the degrees and neighbourhood of the vertices in graph displayed ?



b) Construct the dual graph for the given map. Write the chromatic number for the given graph.



c) Convert the following NFA to DFA. **(5+5+5)**